Welcome.

Everyone:

- Pull the updates from the course GitHub repo:
  - `cd <46120-PiWE repo>`
  - `git pull origin main`

NB:

- By attending this class, you consent to being recorded. Recording will be shared to this class and possibly other DTU students for training purposes.

46120: Scientific Programming for Wind Energy

Midway presentations

Ju Feng

Agenda for today

- Pull new course material ✔

- Round robin
- Reflections and common questions
- Public presentations
- Begin homework

Homework.

- **Homework 0**: Design the architecture of your final project and document it in a diagram. Note that this will be up to changes when you progress more with your final project, don't treat it as final and fixed.
- **Homework 1**: Keep working on your final projects!

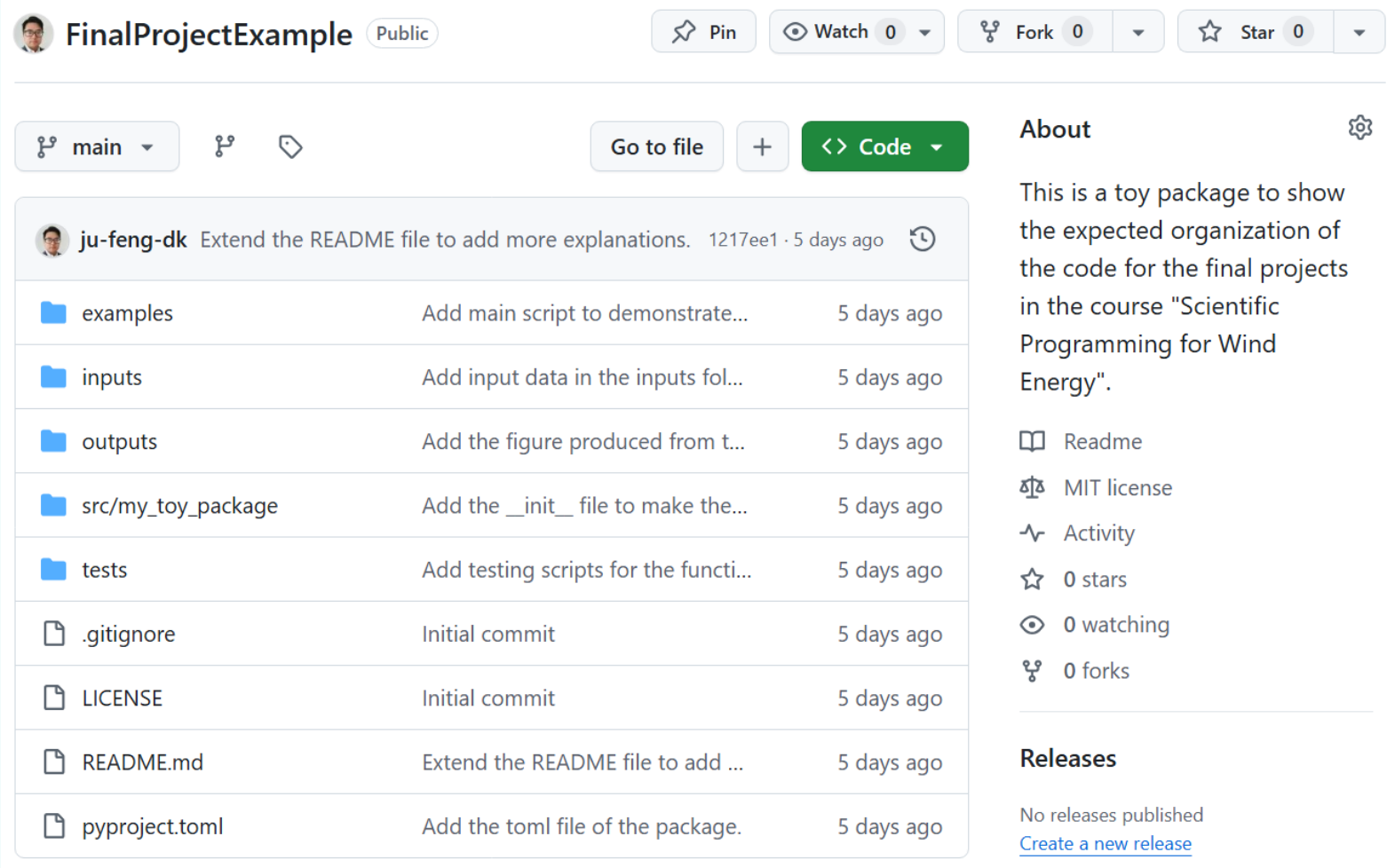
Hand in final project: by Monday, May 4th, at 23:59

A suggested development path

We have a tight schedule, think about your group's development plan. Below is a suggestion to consider:

- **Week 8 and 9**: Understand the project (background, requirements and necessary steps/models/equations), explore the provided data (load, parse, visualize), and consider the preliminary design of the architecture of your package.
- **Week 10**: Implement and refine different parts of the package (including tests).
- **Week 11**: Present your architecture and preliminary results in class, finalize the architecture and polish the code (including tests).
- **Week 12**: Refine the documentation, experiment on different computers (install the package, run the main script, check the test coverage, pylint score), fix all potential issues.

Final project example



https://github.com/ju-feng-dk/FinalProjectExample

Folder names

When you clone the final project template, you may get some folders named differently. **You should change the folder names according to the formal requirements listed in the week 8 README file**, similar to the organization of this example package:

- `inputs` folder: hosts the provided input data, that are needed for your selected final project. Data for other projects should be deleted.
- `examples` folder: hosts the `main.py` script.

Note that the core functions/classes of your package should be defined in the `src/[your_package_name]` folder, and imported in the main script to be used.

Round robin

Share the status of your final project with your peer, discuss about reflections, common challenges and lessons learned.

Time to review and collaborate.

- 1 round of 40 minutes.
- 5 minutes: chaos.
- 35 minutes: present/discuss your final project. Today's feedback focuses:
  1. How is the architecture design?
  2. Are the preliminary results good? Or do you spot any problems?
  3. How is the other team's code? Easy to understand?
  4. Share with other groups how your final project is designed and what is the thinking behind?
  5. What lessons you have learned?
  6. What challenges you are facing?

PLENUM AT 10:00

```
============ Forecasting ============
BOR 1: FreGuLa, EMMUS, Greenland
BOR 2: Pumpkin, ciao_polonia, slumscript
BOR 3: Fasigi, diciassette, Mzansi
BOR 4: TTA, PJ, OmPaDa
BOR 5: Carnami, Group 40, Prompt Engineers
BOR 6: the_best_team, Jarla, JNS
BOR 7: Anjaca, Group_26, Insert_Team_Name_here
BOR 8: ErScPt, HaFlo
BOR 9: Office155, CYWMBW

============ Wind resource assessment ========
BOR 10: Krisacis, Group 10, piwe_cool_kids
BOR 11: PARS, perolacon, camprock-girls
BOR 12: What a Tim, Curry Gang, Pasti_lulus

============ BEM ============
BOR 13: Pasticciotti, Worcestershire, MMC2
BOR 14: Group 6, MaCrIn, Group 19
BOR 15: SSD, PremiumGPT

============ nan ============
BOR 16: Adrion, romaal
```

Notes in plenum (from 2025).

- GET STUFF IN SRC (src/you_package)
- WRITE TESTS AS SOON AS YOU ADD FUNCTIONS/METHODS TO SRC
- MERGE FREQUENTLY
- Recommend: discuss with your team and make a freeze date for src

Reflections and common questions

Public presentations

Volunteer or randomly selected groups. At least one for each final projects.

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BOR 1: FreGuLa, EMMUS, Greenland
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Homework for this week.

*Remember, you're expected to work **about 6 hours outside of class**. Schedule accordingly.*

Homework.

- **Homework 0**: Keep working on your final projects!
  - Work collaboratively with your group.
  - Remember to use branch and PR.
  - Write meaningful commit messages.
  - Check the notes in the folder `notes_and_tips` for inspirations, if you have some challenges in your final project.
  - Check the example package (https://github.com/ju-feng-dk/FinalProjectExample) for organization of the package.

Any questions?

https://github.com/ju-feng-dk/FinalProjectExample